

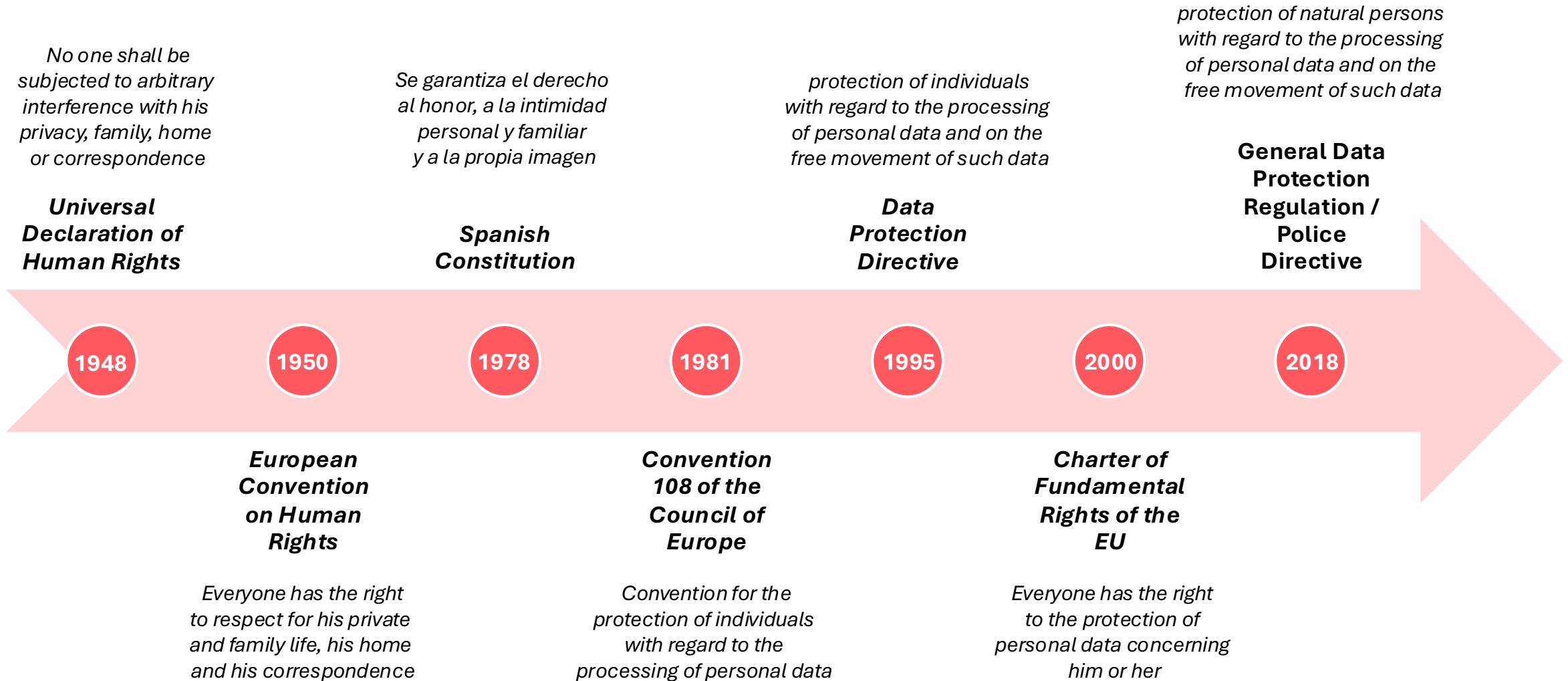
Advancing legal compliance in the Solid ecosystem with DPV

Beatriz Esteves, UGent – imec, BE

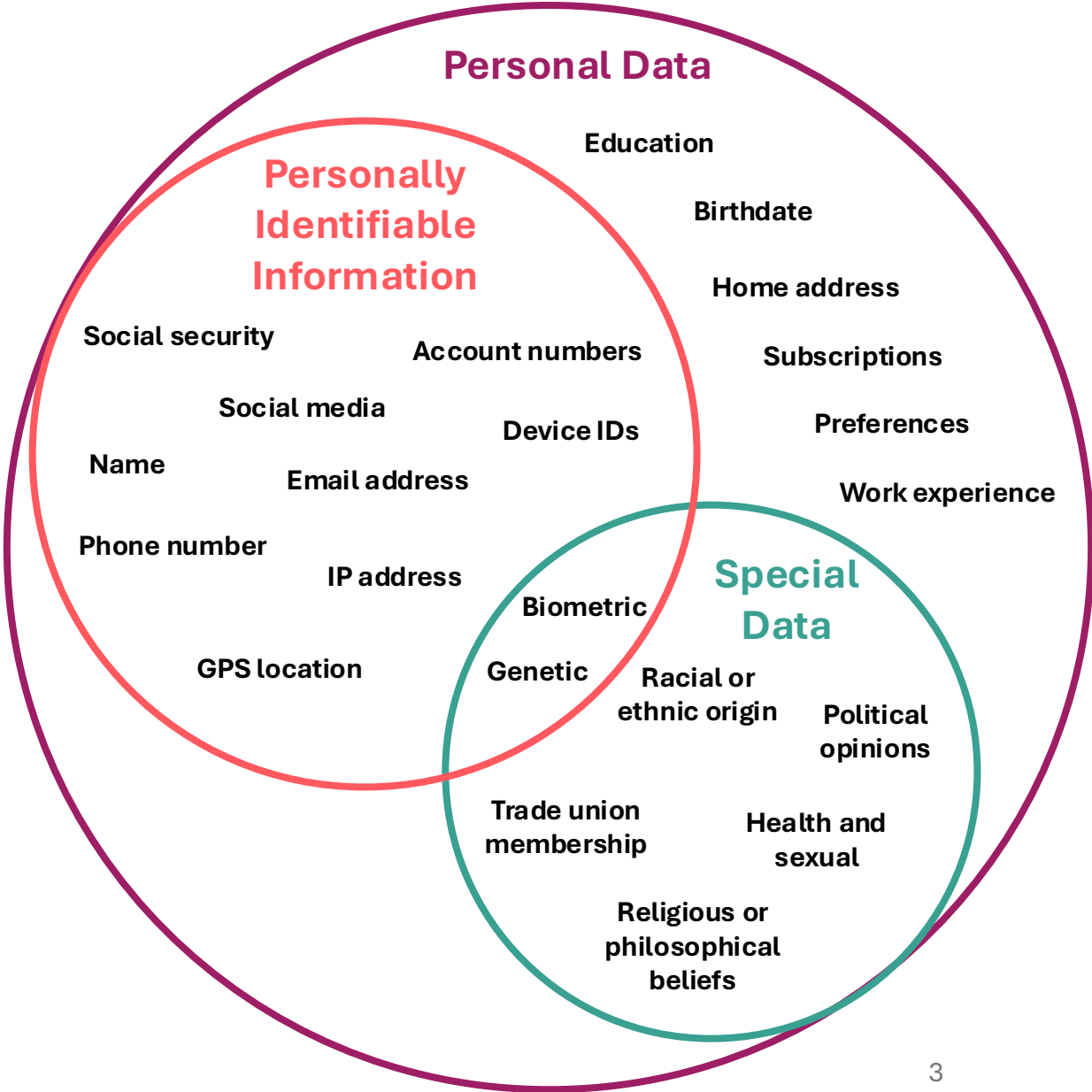
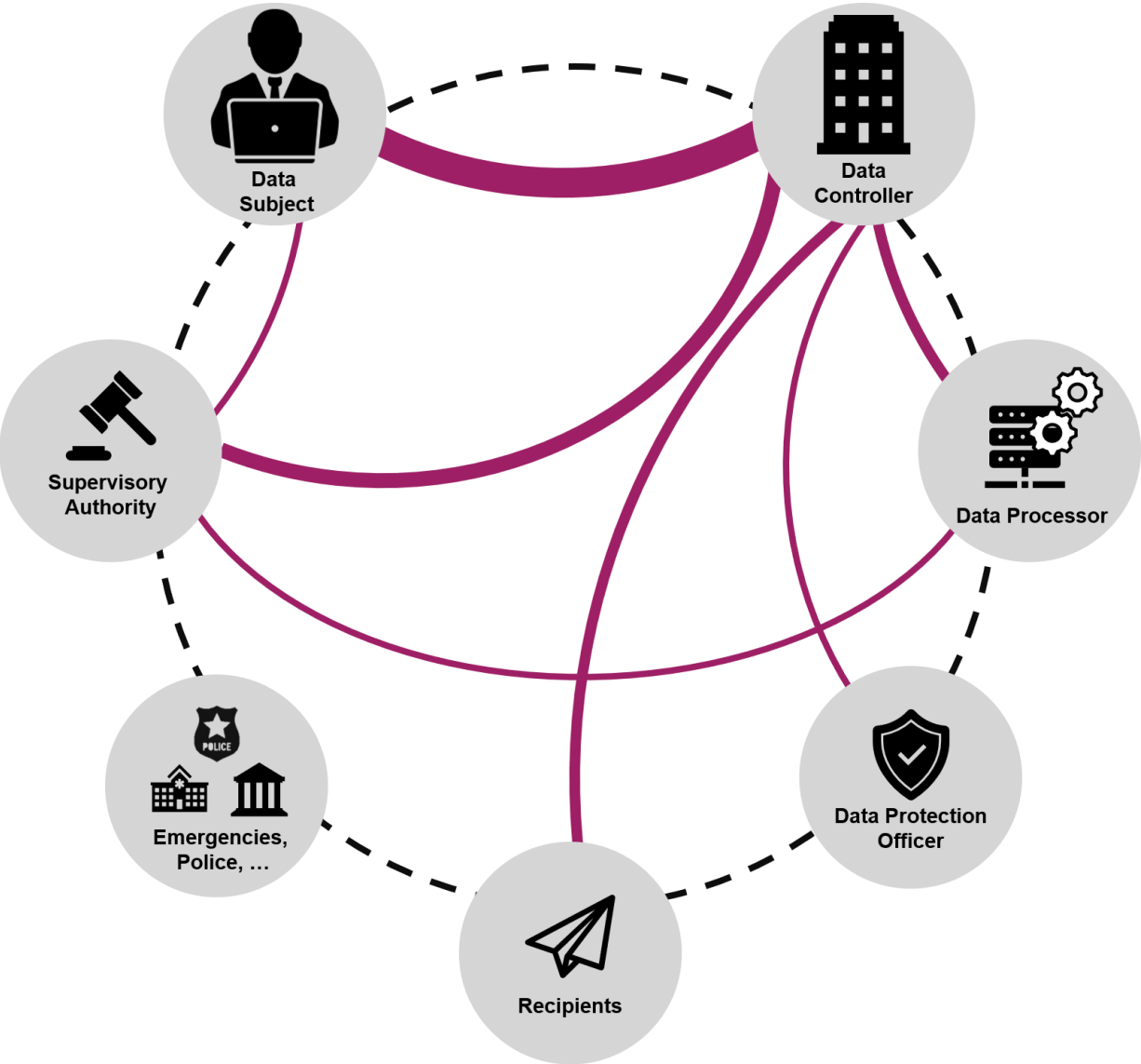
beatriz.esteves@ugent.be



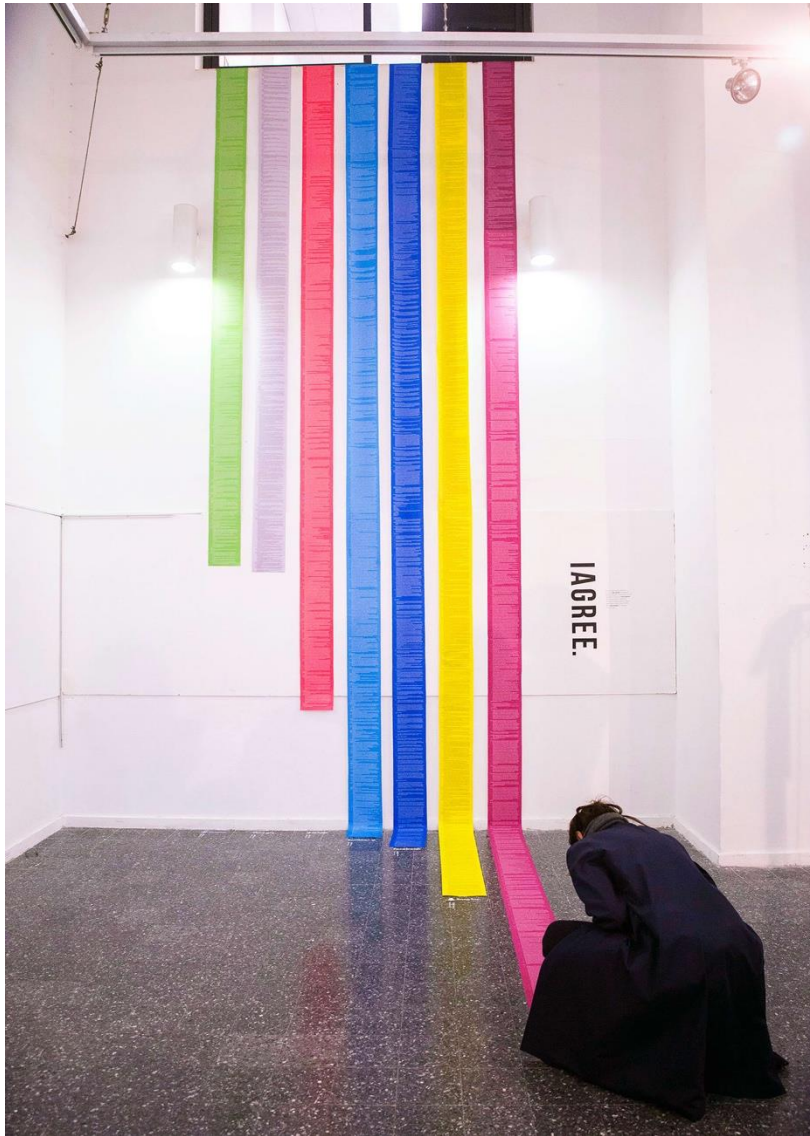
MOTIVATION



DEFINITIONS – GDPR

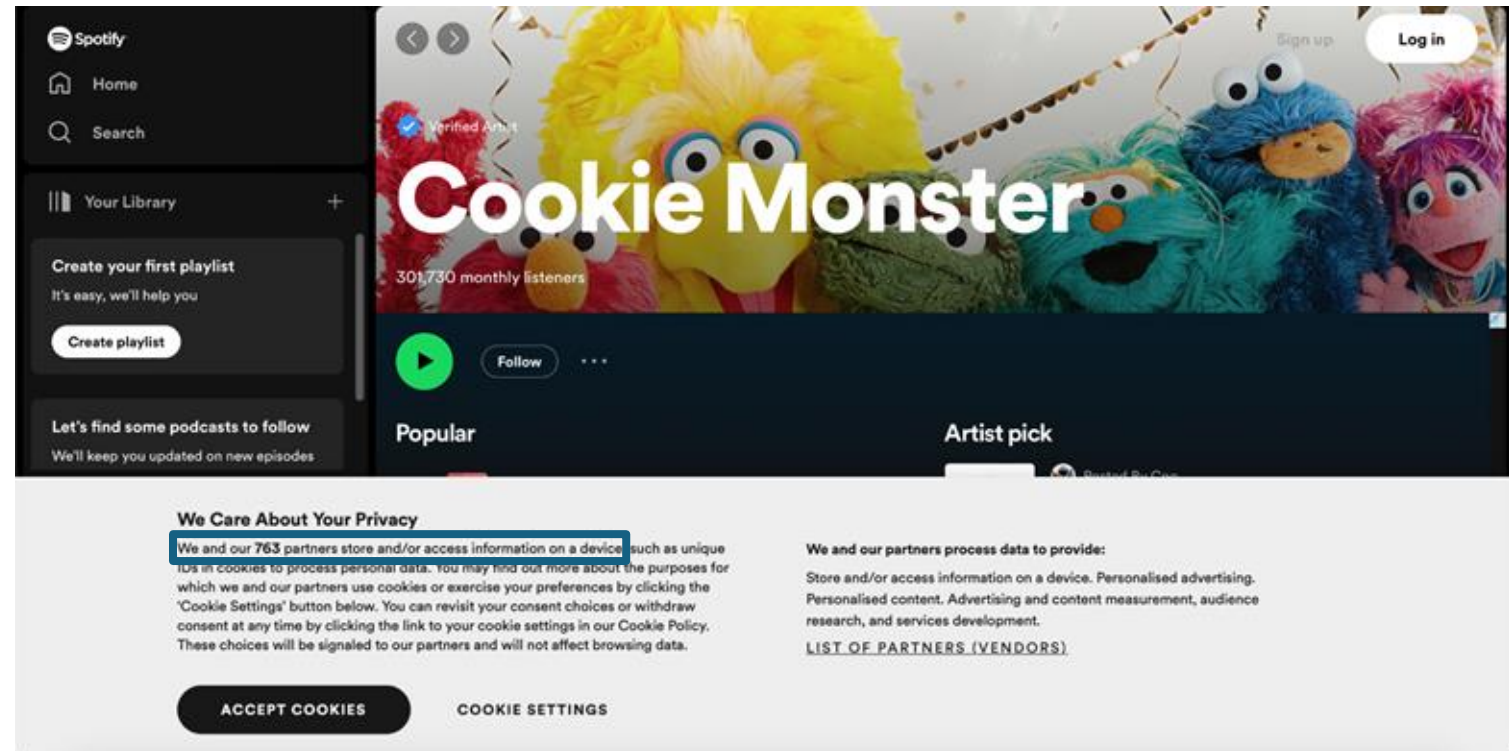


MOTIVATION

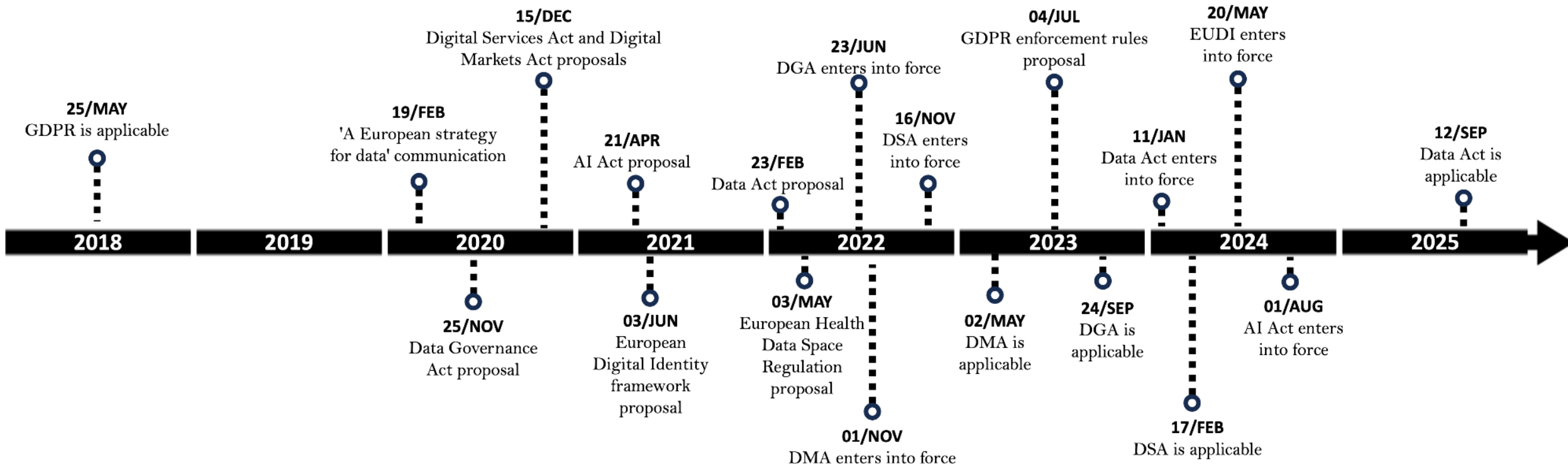


[Photo: Dima Yarovinsky]

<https://www.fastcompany.com/90171107/printing-out-the-privacy-policies-of-facebook-snap-and-others>



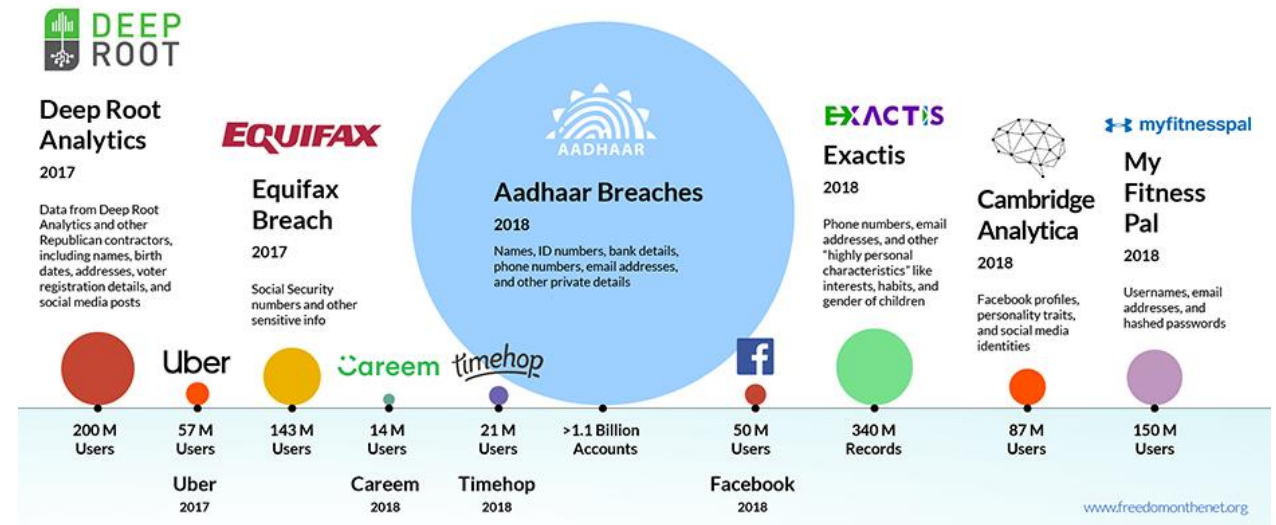
MOTIVATION



MOTIVATION

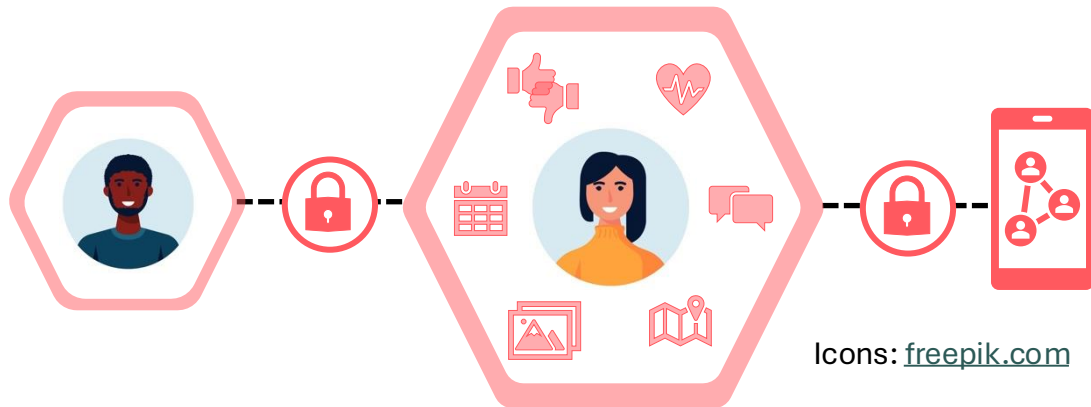


[THE BATTLE OF THE INTERNET: OPEN VS CLOSED – FURTHER EXPLORATION](#), Christopher Langley



DECENTRALISED WEB

solidproject.org



Solid is a specification for decentralised data stores based on interoperable data formats and protocols

Solid's authorisation mechanism currently relies on two access control languages – WAC and ACP

WAC – Web Access Control

```
<#authorization1> a acl:Authorization ;  
  acl:agent <https://beatriz.providerZ.com/profile/card#me> ;  
  acl:accessTo <https://victor.providerY.com/docs/file1.ttl> ;  
  acl:mode acl:Read, acl:Write .
```

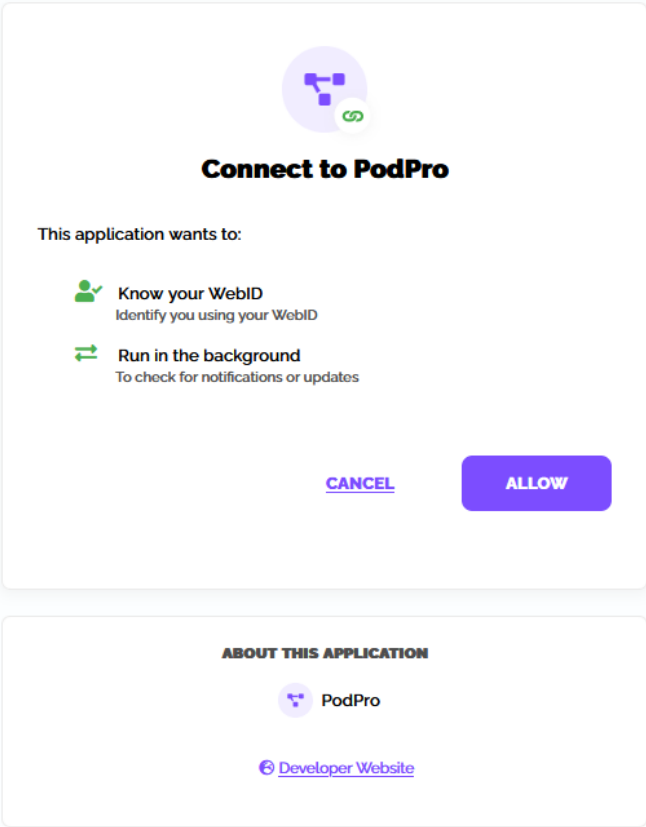
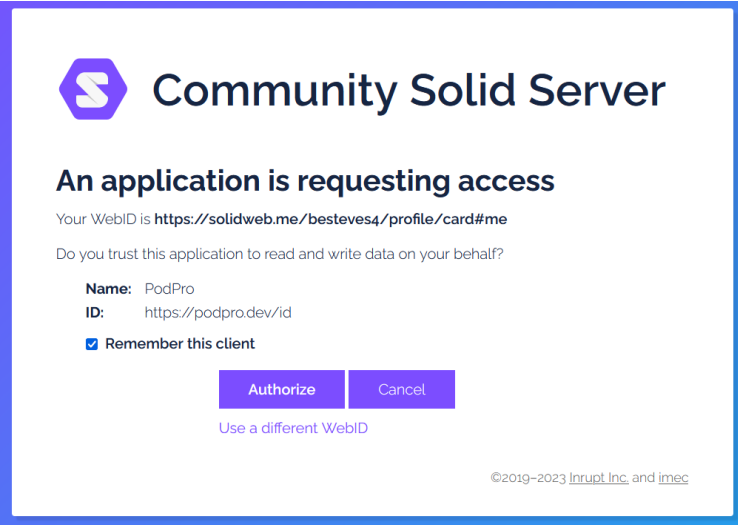
ACP – Access Control Policy

```
<#grant1> a acp:AccessGrant ;  
  acp:grant acl:Read, acl:Write ;  
  acp:context [  
    acp:agent <https://beatriz.providerZ.com/profile/card#me> ;  
    acp:issuer <https://identityProviderZ.com> ;  
    acp:target <https://victor.providerY.com/docs/file1.ttl> ;  
    acp:client <https://clientApplicationA.com> ] .
```

PIMS AS A LEGAL-TECH TOOL

“PIMS provide features for individuals to be able to access their personal data, as well as to rectify or erase them, as provided for by the GDPR, either because the data are in repositories under their direct control or because all shared data are linked to a source, which is again in the control of the individual.”

EDPS TechDispatch #3/2020 – PIMS [Source]



Beatriz Esteves, Haleh Asgarinia, Andres Chomczyk Penedo, Blessing Mutiro, and Dave Lewis. **Fostering trust with transparency in the data economy era: an integrated ethical, legal, and knowledge engineering approach.** In *1st International Workshop on Data Economy*, pages 57–63. ACM, 2022. doi: [10.1145/3565011.3569061](https://doi.org/10.1145/3565011.3569061).



Haleh Asgarinia, Andres Chomczyk Penedo, Beatriz Esteves, and Dave Lewis. **“Who Should I Trust with My Data?” Ethical and Legal Challenges for Innovation in New Decentralized Data Management Technologies.** *Information 14*(7), 2023. doi: [10.3390/info14070351](https://doi.org/10.3390/info14070351).



Marcu Florea and Beatriz Esteves. **Is Automated Consent in Solid GDPR-Compliant? An Approach for Obtaining Valid Consent with the Solid Protocol.** *Information 14*(12), 2023. doi: [10.3390/info14120631](https://doi.org/10.3390/info14120631).

Is this enough for Solid users to know what is happening to their data? Is it enough to comply with GDPR’s requirements?

DECENTRALISED WEB

References	GDPR	Data Subject Rights							Principles Art.5							
		Portability	Withdraw consent	Access	Rectification	Forgotten	Notification	Object	Automated Decision- Making	Lawfulness, fairness, transparency	Purpose limitation	Data minimisation	Accuracy	Storage limitation	Integrity and Confidentiality	Accountability
Byule et al. 2020		●										●				
Wang 2020				●	●					●		●				
Ammar et al. 2021											▲					
De Bot and Haegemans 2021										●	●					●
De Mulder et al. 2021		✕								✕						
Janeiro Digital 2021										▲				▲		
PDS Interop 2021		✕														
Tóth 2022					◆	◆										
Van Damme 2022		✕								✕						✕
Van de Wynckel and Signer 2022										★						
Verstraete et al. 2022			■			■										
Bailly et all. 2023										✕						
Esposito et al. 2023							▲	▲	▲			▲			▲	▲
Pandit 2023			✕							✕	✕	✕	✕	✕	✕	✕
Sun et al. 2023			▲		▲					▲	▲	▲		▲	▲	

Red – theory

Black – apps

Orange – identity provider

Blue – Pod provider

○ – Government

△ – Health

☆ – Location

□ – Human resources

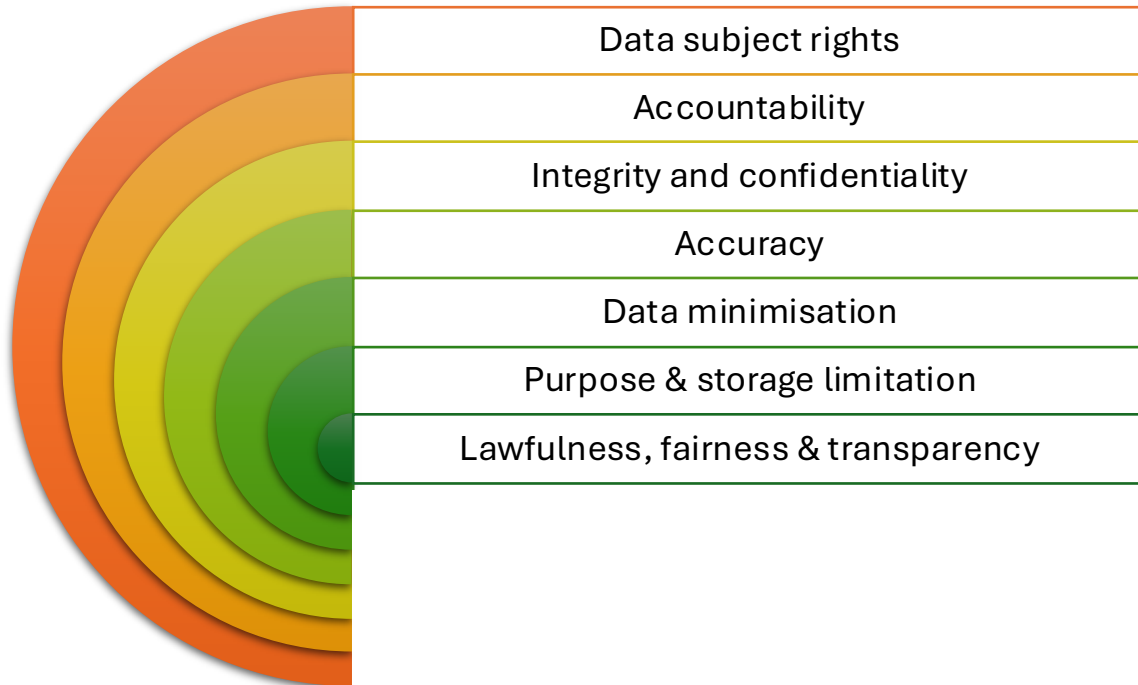
◇ – Hospitality

✕ – No domain

- Strong focus on the ‘**lawfulness, fairness and transparency**’ principle
- Distinct works were also found to tackle the right to **data portability, withdrawal of consent**, and **rectification**
- No work was found on the right to **restrict the processing** of personal data

“PIMS provide features for individuals to be able to access their personal data, as well as to rectify or erase them, as provided for by the GDPR, either because the data are in repositories under their direct control or because all shared data are linked to a source, which is again in the control of the individual.”

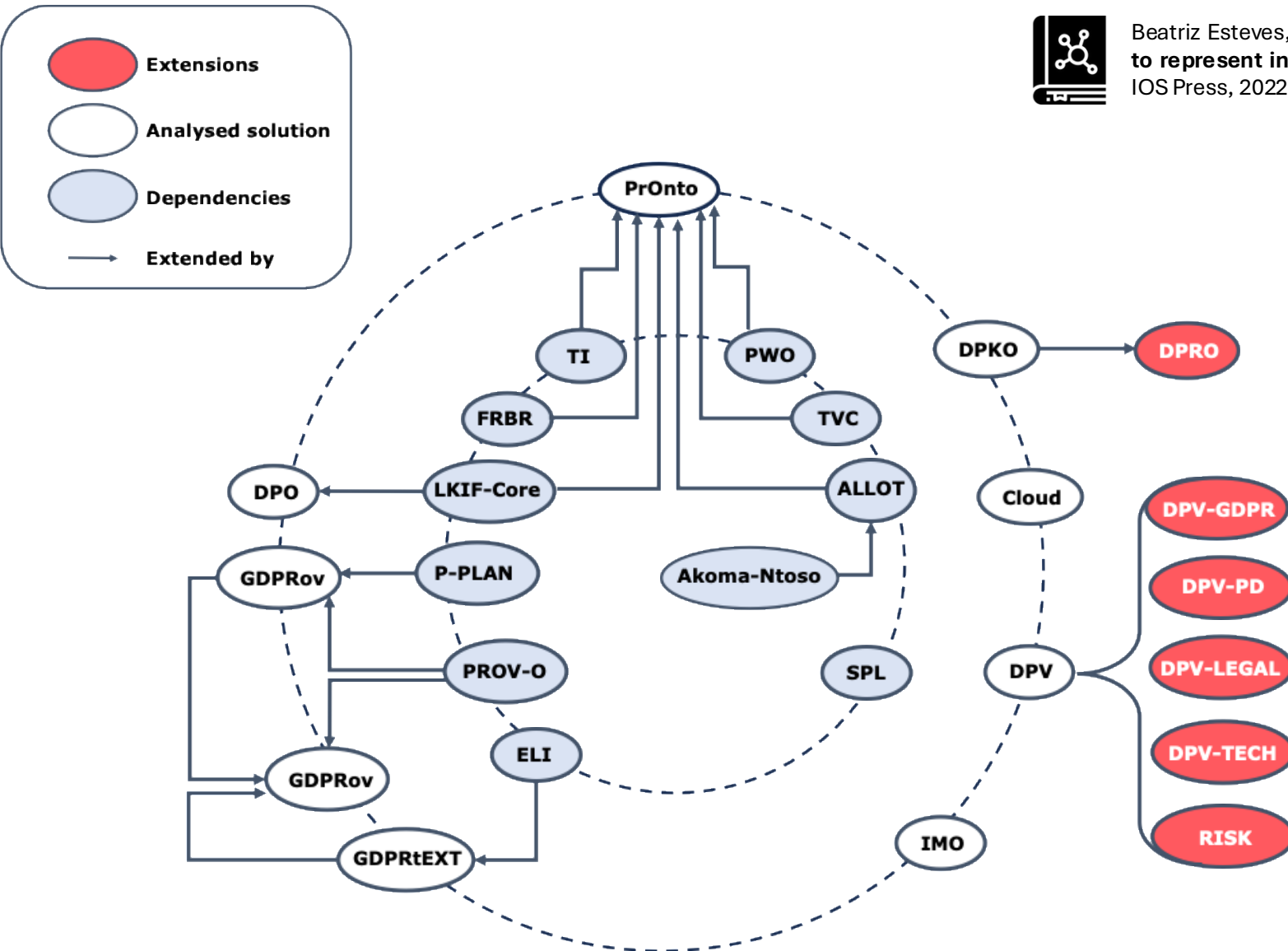
EDPS TechDispatch #3/2020 – PIMS [\[Source\]](#)



What is missing from Solid?

- No records of agreements for the provision of services
- Lack of tools to give / withdraw consent
- No metadata about Solid infrastructure
- No information on the identity and contacts of Solid providers
- Lack of tools to get available categories of data
- Record keeping and log maintenance are nonexistent
- Lack of tools to rectify data inaccuracies
- Difficulty for users to set (granular) access to resources
- Access grants valid in perpetuity
- Data requests miss a purpose
- Compatibility of purposes cannot be checked
- Consent dialogue not enough for informed decision
- Access grants not sufficient to be a valid record of consent

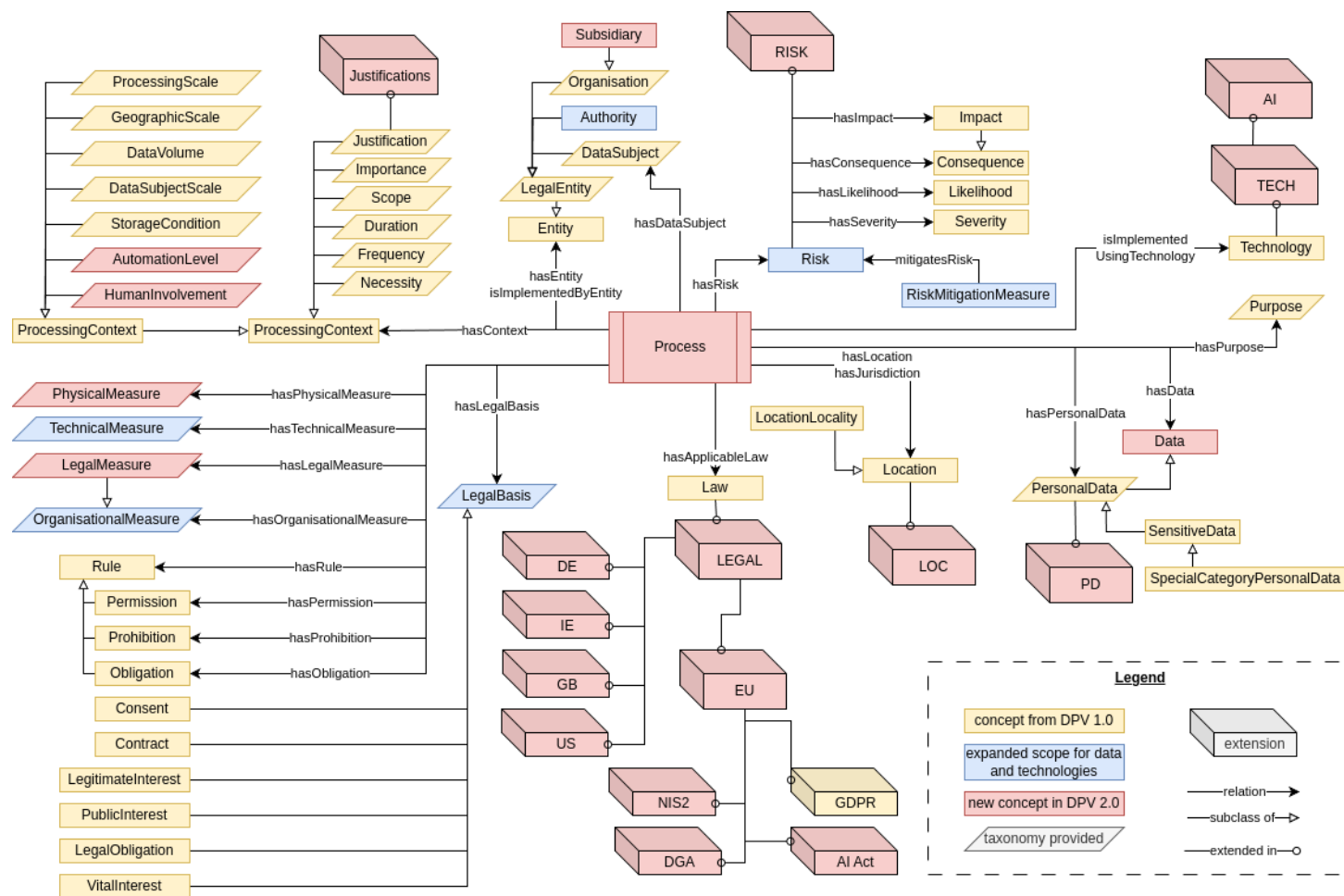
DATA PROTECTION VOCABULARIES



Beatriz Esteves, Victor Rodriguez-Doncel, **Analysis of ontologies and policy languages to represent information flows in GDPR**. *Semantic Web Journal* 15(3), pages 709-743. IOS Press, 2022. doi: [10.3233/SW-223009](https://doi.org/10.3233/SW-223009).

- **DPV** and **GDPRtEXT** include the most concepts to represent, at least partially, privacy terms from the ‘right to be informed’ (Arts. 13 and 14) and other data subject rights (Arts. 15 to 22)
- Only DPV has been updated in the past two years
- DPKO, IMO, and PrOnto lack open and accessible resources

DATA PRIVACY VOCABULARY



Data Privacy Vocabulary (DPV)

<https://w3id.org/dpv>

- Developed by the **W3C** Data Privacy Vocabularies and Controls Community Group (**DPVCG**)
- Defines a **jurisdiction-agnostic** ontology for expressing metadata about the processing of personal data
- Provides **hierarchical taxonomies**, from abstract to more specific concepts, to instantiate specific concepts in practical use-cases
- Has law-specific extensions

Harshvardhan J. Pandit, Axel Polleres, Bert Bos, Rob Brennan, Bud Bruegger, Fajar J. Ekaputra, Javier D. Fernández, Roghaiyeh Gachpaz Hamed, Elmar Kiesling, Mark Lizar, Eva Schlehahn, Simon Steyskal, and Rigo Wenning. **Creating a Vocabulary for Data Privacy: The First-Year Report of Data Privacy Vocabularies and Controls Community Group (DPVCG)**. *On the Move to Meaningful Internet Systems: OTM 2019 Conferences*, volume 11877 of Lecture Notes in Computer Science, pages 714–730. Springer International Publishing, 2019. doi: [10.1007/978-3-030-33246-4_44](https://doi.org/10.1007/978-3-030-33246-4_44).

Harshvardhan J. Pandit, Beatriz Esteves, Georg P. Krog, Paul Ryan, Delaram Golpayegani, and Julian Flake. **Data Privacy Vocabulary (DPV) – Version 2.0**. In *The Semantic Web – ISWC 2024*, pages 171–93. Cham: Springer Nature Switzerland, 2024. doi: [10.1007/978-3-031-77847-6_10](https://doi.org/10.1007/978-3-031-77847-6_10).

DATA PRIVACY VOCABULARY

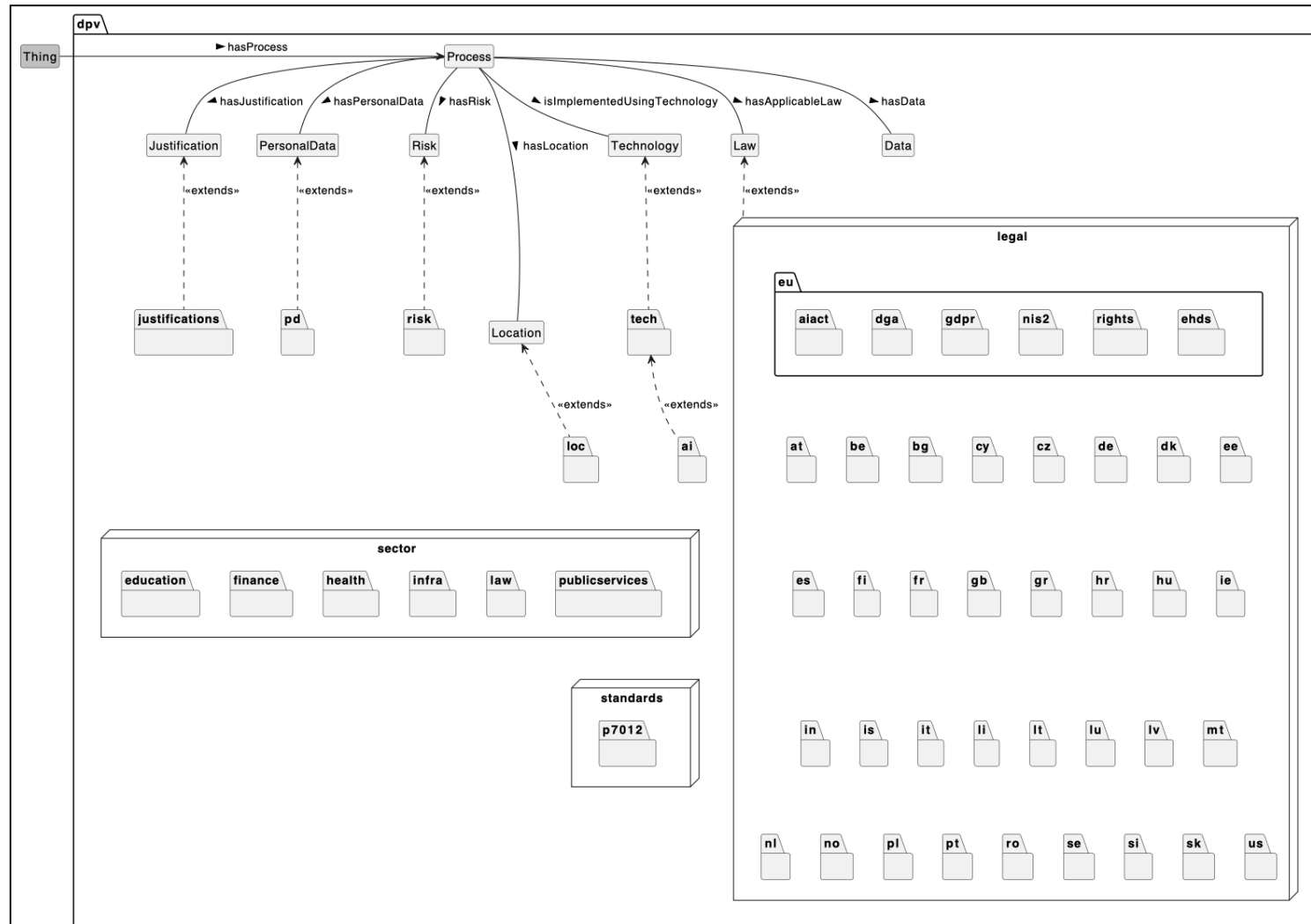


Figure 20 Structure of DPV vocabularies where DPV defines the core concepts which are then extended in specific extensions. The LEGAL extensions are named using ISO 3166-2 country codes, and contain specific extensions modelling laws within that jurisdiction. SECTOR and STANDARDS extensions also contain extensions within them modelling specific sectors and standards respectively.

GENERAL DATA PROTECTION REGULATION

 W3C Community Group
Final Report

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- 2. Mapping GDPR concepts to DPV
 - 3. Legal Basis
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 - 3.3 Data Transfer (Art.45-49)
 - 4. Principles
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 - 5.3 Mapping: Legal Basis x Rights
 - 5.4 Justifications for Rights Exercise
 - 6. Data Transfer Tools

EU General Data Protection Regulation (GDPR)

version 2.1

Final Community Group Report 16 March 2025

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<https://www.w3.org/community/reports/dpvcg/CG-FINAL-eu-gdpr-20250316/>

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Beatriz Esteves  (IDLab, IMEC, Ghent University)
Georg P. Krog (Signatu AS)
Harshvardhan J. Pandit  (ADAPT Centre, Dublin City University)
Paul Ryan  (Uniphar PLC)

Feedback:

[GitHub w3c/dpv](https://github.com/w3c/dpv) (pull requests, new issue, open issues)

This Release

<https://w3id.org/dpv/2.1/legal/eu/gdpr>

Previous Release

<https://w3id.org/dpv/2.0/legal/eu/gdpr>

Changelog

[Changelog for v2.1](#)

Key Publications

Data Privacy Vocabulary (DPV) -- Version 2.0 (2024)

 W3C Community Group
Final Report

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- Status of This Document
- 1. Primer introducing DPV
- 2. Guides for using DPV concepts
 - 2.1 Using DPV with RDFS and SKOS
 - 2.2 Using DPV with OWL2
 - 2.3 Using DPV with JSON, CSV, etc.
 - 2.4 Using DPV with ODRL
- 3. Guides for DPV Applications
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 - 3.2 ISO-29184 Privacy Notices
 - 3.3 Record of Processing Activities (ROPA)
 - 3.4 Data Protection Impact Assessment (DPIA)
 - 3.5 Data Breach Records and Notifications
 - 3.6 Rights Management
- 4. Use-Cases, Requirements, and Examples

Guides for Data Privacy Vocabulary (DPV)

Final Community Group Report 16 January 2025

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[GitHub w3c/dpv](https://github.com/w3c/dpv) (pull requests, new issue, open issues)

Key Publications

Data Privacy Vocabulary (DPV) -- Version 2.0 (2024)

Copyright © 2025 the Contributors to the Guides for Data Privacy Vocabulary (DPV) Specification, published by the Data Privacy Vocabularies and Controls Community Group under the W3C Community Final Specification Agreement (FSA). A human-readable summary is available.

Abstract

This document lists the various guides created by the DPVCG and the community providing guidance for the adoption and use of DPV in terms of its concepts and serialisations, or regarding the application of DPV for specific applications or domains.

The DPVCG invites contributions regarding additional guides as well as updates to existing guides.

<https://w3id.org/dpv/legal/eu/gdpr>

<https://w3id.org/dpv/guides>



Harshvardhan J. Pandit, Beatriz Esteves, Georg P. Krog, Paul Ryan, Delaram Golpayegani, and Julian Flake. **Data Privacy Vocabulary (DPV) – Version 2.0**. In *The Semantic Web – ISWC 2024*, pages 171–93. Cham: Springer Nature Switzerland, 2024. doi: [10.1007/978-3-031-77847-6_10](https://doi.org/10.1007/978-3-031-77847-6_10).



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3.1.8 Crime Detection Impaired

3.1.9 Crime Investigation Impaired

Justifications Extension

version 2.1

Final Community Group Report 16 March 2025

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Authors:

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Harshvardhan J. Pandit (ADAPT Centre, Dublin City University)
Paul Ryan (Uniphar PLC)

Feedback:

GitHub w3c/dpv (pull requests, new issue, open issues)

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<https://w3id.org/dpv/2.1/justifications>

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<https://w3id.org/dpv/2.0/justifications>

Changelog

Changelog for v2.1

Key Publications

Beatriz Esteves, Victor Rodriguez-Doncel, and Ricardo Longares. *Automating the Response to GDPR's Right of Access*. In *Legal Knowledge and Information Systems*, pages 170–175. IOS Press, 2022. doi: [10.3233/FAIA220462](https://doi.org/10.3233/FAIA220462).

<https://w3id.org/dpv/justifications>

<https://w3id.org/dpv/guides/rights>

```
ex:RejectRightToErasure a dpv:RightNonFulfilmentNotice ;
  dcterms:issued "2024-09-06"^^xsd:date ;
  dcterms:description "Notice of non-fulfillment related to an exercised right to erasure" ;
  dcterms:identifier "x4ghyun-658393" ;
  dcterms:language "EN" ;
  dcterms:publisher ex:DataController ;
  dpv:hasRight eu-gdpr:A17 ;
  dpv:hasDataController ex:DataController ;
  dpv:isImplementedByEntity ex:DataController ;
  foaf:page <https://example.org/DataController/RejectRightToErasure> ;
  dpv:hasRecipient ex:DataSubject ;
  dpv:hasStatus dpv:RequestUnfulfilled ;
  dpv:hasJustification justifications:FreedomOfExpressionImpaired .
```

```
ex:DataSubject a dpv:DataSubject .
```



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A.1 Informative references

UNOFFICIAL

Guide for using DPV for Rights Exercise and Management

Work in Progress

Draft Community Group Report 16 January 2025

Latest published version:

<https://w3id.org/dpv/guides/rights>

Latest editor's draft:

<https://dev.dpvcg.org/guides/rights>

Editor:

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Feedback:

GitHub w3c/dpv (pull requests, new issue, open issues)

Key Publications

Data Privacy Vocabulary (DPV) -- Version 2.0 (2024)

Copyright © 2025 the Contributors to the Guide for using DPV for Rights Exercise and Management Specification, published by the Data Privacy Vocabulary and Controls Community Group under the W3C Community Contributor License Agreement (CLA). A human-readable summary is available.

Abstract

This document will provide a guide for using DPV for Rights Exercise and Management. Currently, it is a work in progress.

DPV Specifications: The [DPV] is the core specification within the DPV family, with the following extensions: Personal Data [PD], Locations [LOC], Risk Management [RISK], Technology [TECH] and [AI].



Beatriz Esteves, Victor Rodriguez-Doncel, and Ricardo Longares. **Automating the Response to GDPR's Right of Access**. In *Legal Knowledge and Information Systems*, pages 170–175. IOS Press, 2022. doi: [10.3233/FAIA220462](https://doi.org/10.3233/FAIA220462).



Beatriz Esteves, Harshvardhan J. Pandit, Georg P. Krog, and Paul Ryan. **How to Manage My Data? With Machine-Interpretable GDPR Rights!** In *Legal Knowledge and Information Systems*, 269–74. IOS Press, 2024. doi: [10.3233/FAIA241254](https://doi.org/10.3233/FAIA241254).

DATA GOVERNANCE ACT



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EU Data Governance Act (DGA)

version 2.1

[Final Community Group Report 16 March 2025](#)

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Feedback:

[GitHub w3c/dpv](#) ([pull requests](#), [new issue](#), [open issues](#))

This Release

<https://w3id.org/dpv/2.1/legal/eu/dga>

Previous Release

<https://w3id.org/dpv/2.0/legal/eu/dga>

Changelog

[Changelog for v2.1](#)

Key Publications

[Data Privacy Vocabulary \(DPV\) -- Version 4](#)

Legal terms validated by legal scholars and Data Protection Officers and integrated into the outcomes of the W3C DPVCG

Use Cases

UC1. Conditions for the Reuse of Public Data

UC2. Policies for Data Altruism

UC3. Records of Altruistic and Intermediation Activities

UC4. Registers of Entities

<https://w3id.org/dpv/legal/eu/dga>



Beatriz Esteves and Víctor Rodríguez-Doncel. **Semantifying the Governance of Data in Europe**. In *18th International Conference on Semantic Systems – CEUR Workshop Proceedings*, volume 3235, 2022. URL: <https://ceur-ws.org/Vol-3235/paper17.pdf>.



Beatriz Esteves, Víctor Rodríguez-Doncel, Harshvardhan J. Pandit, and Dave Lewis. **Semantics for Implementing Data Reuse and Altruism Under EU's Data Governance Act**. In *Knowledge Graphs: Semantics, Machine Learning, and Languages*, pages 210–226. IOS Press, 2023. doi: [10.3233/SSW230015](https://doi.org/10.3233/SSW230015).

DATA GOVERNANCE ACT

```
ex:SIPPA_assets a :DataAssetList, dcat:Catalog ;
  dct:description "Asset list maintained by SIPPA" ;
  dct:created "2022-12-10"^^xsd:date ;
  dct:publisher ex:SIPPA ; dcat:dataset ex:dataset_001 .
ex:SIPPA a :SingleInformationPointProvider .
ex:dataset_001 a dcat:Dataset ; dct:publisher ex:publicsectorbodyX ;
  dpv:hasData :StatisticallyConfidentialData ;
  dct:description "Dataset with statistically confidential data" ;
  dct:created "2022-12-04"^^xsd:date ;
  odrl:hasPolicy ex:policy_001 ; :hasFee "0€"^^xsd:string ;
  dcat:mediaType <iana.org/assignments/media-types/text/csv> ;
  dct:extent "5.6MB"^^xsd:string .
```

```
ex:policy_001 a odrl:Offer, :DataReusePolicy ;
  odrl:permission [
    odrl:target ex:dataset_001 ; odrl:action :Reuse ;
    odrl:assigner ex:publicsectorbodyX ;
    odrl:constraint [
      odrl:and [
        odrl:leftOperand odrl:dateTime ;
        odrl:operator odrl:lteq ;
        odrl:rightOperand "2023-12-31"^^xsd:date ] , [
        odrl:leftOperand odrl:purpose ;
        odrl:operator odrl:isA ;
        odrl:rightOperand :ScientificResearch ] ] ] .
ex:publicsectorbodyX a :PublicSectorBody ;
  dpv:hasName "Public Sector Body X" ;
  dpv:hasContact "mailto:publicsectorbodyX@email.com" ;
  :hasCompetentBody [
    a :DataReuseCompetentBody ; dpv:hasName "Competent Body X" ;
    dpv:hasContact "mailto:competentbodyX@email.com" ] .
```

```
ex:publicregistry_DI_PT a :RegisterOfDataIntermediationServiceProviders ;
  dct:description "Public register of intermediaries working in PT" ;
  dct:created "2023-12-15"^^xsd:date ;
  dct:modified "2023-12-23"^^xsd:date ;
  dct:publisher ex:nationalauthority_PT ;
  :hasDataIntermediationServiceProvider ex:DISP_Y .
ex:nationalauthority_PT a :DataIntermediationAuthority ;
  dpv:hasName "Data Intermediation Authority of Portugal" ;
  dpv:hasContact "mailto:nationalauthority_PT@email.com" ;
  dpv:hasJurisdiction "PT" .
ex:DISP_Y a :DataCooperative ;
  dpv:hasName "Data Cooperative Y" ; dpv:hasAddress "Lisboa, Portugal" ;
  dct:description "Provider of anonymised geolocation data" ;
  dcat:landingPage <http://cooperativeA.com/> ;
  dct:date "2023-12-23"^^xsd:date .
```

```
ex:altruism_logs a :RegisterOfDataAltruismActivity ;
  dct:description "Activity logs of the Data Altruism Organisation A" ;
  dct:created "2023-11-04"^^xsd:date ;
  dct:modified "2023-11-13"^^xsd:date ;
  dct:publisher ex:altruism_A ; dcat:record ex:log_001 .
ex:altruism_A a :DataAltruismOrganisation ;
  dpv:hasName "Data Altruism Organisation A" ;
  dpv:hasAddress "Lisboa, Portugal" ;
  dcat:landingPage <http://example.com/altruism_A> .
ex:log_001 a dcat:CatalogRecord ;
  dct:created "2023-11-13"^^xsd:date ;
  :hasDataUser ex:userZ ; :hasFee "1000€"^^xsd:string ;
  dpv:hasPersonalDataHandling [
    dct:description "Download and reuse anonymised health records to
    ↪ improve healthcare" ;
    dpv:hasProcessing :Download, :Reuse ; dpv:hasDuration 6226453 ;
    dpv:hasPurpose :DataAltruism, :ImproveHealthcare ;
    dpv:hasPersonalData dpv-pd:HealthRecord ;
    dpv:hasTechnicalMeasure dpv:Anonymisation ] .
ex:userZ a :DataUser ; dpv:hasName "Data User Z" ;
  dpv:hasContact "mailto:user_z@email.com" .
```



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Abstract

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EU European Health Data Spaces (EHDS)

version 2.1

Draft Community Group Report 16 March 2025

Latest published version:

<https://w3id.org/dpv/legal/eu/ehds>

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Feedback:

[GitHub w3c/dpv](#) ([pull requests](#), [new issue](#), [open issues](#))

This Release

<https://w3id.org/dpv/2.1/legal/eu/ehds>

Key Publications

[Data Privacy Vocabulary \(DPV\) -- Version 2.0 \(2024\)](#)

[Copyright](#) © 2025 the Contributors to the EU European Health Data Spaces (EHDS) Specification, published by the [Data Privacy Vocabularies and Controls Community Group](#) under the [W3C Community Contributor License Agreement \(CLA\)](#). A human-readable [summary](#) is available.

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WHO IS USING DPV? WHAT ARE THEY DOING?

Year	Mention	Use	Ext.	Contrib.	Domain	Effort
2020	X				Health	N/A
2020	X				Media	N/A
2020	X					N/A
2020		X				++
2020		X			Health	++
2020			X	X		++
2021	X				Health	N/A
2021	X					N/A
2021		X				++
2021		X			Smart products	+
2021			X	X		+
2022	X					N/A
2022	X			X		N/A
2022			X	X		+
2022	X				Health	N/A
2022		X				+
2022		X			IoT	+
2022		X			Health	+

Creating a **vocabulary** for data privacy: The first-year report of **data privacy vocabularies** and controls community group (DPVCG)
HJ Pandit, A Polleres, B Bos, R Brennan... - On the Move to ..., 2019 - Springer
...) was set up to jointly develop such **vocabularies** towards interoperability in the context of **data privacy**. This paper presents the resulting **Data Privacy Vocabulary** (DPV), along with a ...
☆ Save  Cite **Cited by 96** Related articles All 9 versions

(At the time of writing the paper we had 81 citations)

We looked at how much *Effort* it would take for these works to integrate the changes made in DPV.

Most works only required minor changes in changed concepts (+ *in table*), other works required changing IRIs to the new formats as they were using an old pre-v1.0 draft version of the DPV.

🏆 Prototyping an End-User User Interface for the Solid Application Interoperability Specification under GDPR by H Bailly, A Papanna, R Brennan wins **Best In-Use Paper award** at ESWC 2023 🏆

2023		X			Smart cities	+
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🏆 Using Patterns to Manage Governance of Solid Apps by B. Esteves, H. Pandit wins **Best Paper award** at Workshop on Ontology Patterns (WOP), ISWC 2023 🏆

2024		X			...	...
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🏆 Implementing ISO/IEC TS 27560: 2023 Consent Records and Receipts for GDPR and DGA by H. Pandit, J. Lindquist, G. P. Krog wins **Best Paper award** at Annual Privacy Forum 2024 organised by European Union Agency for Cybersecurity (ENISA) and EU Commission 🏆

WHO IS USING DPV? WHAT ARE THEY DOING?

Creating a vocabulary for data privacy: The first-year report of data privacy vocabularies and controls community group (DPVCG) HJ Pandit, A Polleres, B Bos, R Brennan, B Bruegger, FJ Ekaputra, ... OTM Confederated International Conferences" On the Move to Meaningful ...	114	2019
Data Privacy Vocabulary (DPV)–Version 2.0 H J. Pandit, B Esteves, G P. Krog, P Ryan, D Golpayegani, J Flake International Semantic Web Conference, 171-193	46	2024

WHAT PROJECTS AND INDUSTRIES ARE USING DPV?

EU Horizon Projects

SPECIAL TRAPEZE

MOSAICrOWN

smashHit FAIRVASC

PROTECT ITN

Industry

Signatu Inrupt JLINC

Athumi / Solid Lab

* These are the projects we know about. Most uses of DPV go unreported, and DPV is also commonly referenced in footnotes and inline text which we do not have a way to track. Projects also contained industry partners.

Spotlight: Signatu's use of DPV (Georg P. Krog)

Signatu: Practical Implementation

Signatu provides a practical solution for implementing ISO/IEC TS 27560:2023 using the the Data Privacy Taxonomy for GDPR and DGA compliance. Signatu's Legal Compliance and Consent Management Platforms offer tools for creating, managing, and exchanging consent records and receipts in line with the latest standards.

Here's an overview of the Data Processing Specification for the "Email Newsletter":

- **Name:** Email Newsletter
- **Purpose:** To distribute regular updates, promotions, and information to subscribers via email.
- **Purpose Category:** dpv:DirectMarketing
- **Personal Data Collected:**
 - Email address
 - Name (optional)

Legal Basis: Consent

Now, let's map the personal data to the DPV personal data categories:

- **Email address** corresponds to `pd:Email Address`
- **Name** corresponds to `pd:Name`

Would you like to proceed with this overview, or do you have any changes?

Description of Data *

Email address, Name

Data Categories

Email Address ☐ Name ☐

Email Address

dpv:PersonalData / pd:Email Address

Information about Email address.

Name

dpv:PersonalData / pd:Name

Data Entry Chatbot

Taxonomy Representation

To distribute regular updates, promotions, and information to subscribers via email. ☒

Detailed information and settings

What is the purpose of collecting data?	Which data do we collect?	What is the legal basis of the purpose?	What is the benefit for you?	Who receives the data?
To distribute regular updates, promotions, and information to subscribers	Email address, Name	Art 6(1) consent	Receive newsletter	MailChimp ☒ Read more

Generate Consent Notice

```
"dpv:hasProcess": [ 1 item
  0: { 3 items
    "@type": "dpv:DecisionProcess"
    "dpv:hasRecipient": [ 1 item
      0: string "company:657"
    ]
    "dpv:hasConsentStatus": [ 1 item
      0: { 2 items
        "@type": [ 2 items
          0: string "dpv:ConsentRefused"
          1: string "dpv:ExpressedConsent"
        ]
        "dpv:isIndicatedAtTime": string "2024-10-16T13:07:12."
```

Generate Consent Records with Events

https://www.w3.org/community/dpvcg/wiki/Adoption_of_DPVCG





Come join us in the W3C Data Privacy Vocabularies and Controls Community Group (DPVCG) !!!

Find a short intro at <https://www.dpvcg.org/>

Primer

Data Privacy Vocabulary (DPV)

[Final Community Group Report 01 August 2024](#)

<https://w3id.org/dpv/primer>



IEEE P7012

IEEE Draft Standard for Machine Readable Personal Privacy Terms

IEEE P7012 is using DPV to express terms

More Jurisdictions

AI taxonomies

ODRL alignment

Risks/Impacts

AI Act compliance

Guidance for using
DCAT and PROV

New ISO standard proposed
based on DPV modelling of
Legal Bases

Machine-Actionable Rights



~ NXDG 2025 ~

NeXt-generation Data Governance workshop
2025

co-located with **21st SEMANTiCS**
3-5 September 2025
Vienna, Austria

<https://w3id.org/nxdg/2025>



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Advancing legal compliance in the Solid ecosystem with DPV

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